

the average execution price and the price of the stock at the time the order was entered into the market (arrival price). A positive value indicates more favorable transaction prices and a negative value indicates less favorable transaction prices. Trading PnL is a measure of the cost during trading and reports whether the investor did better or worse than the arrival price. For example, a trading PnL of -10 bp indicates the fund underperformed the arrival price benchmark by 10 bp. The formulation for trading PnL multiplies the arrival cost calculation above by minus 1. This is:

$$\text{Trading PnL}_{bp} = -1 \cdot \text{Side} \cdot \frac{P_{avg} - P_0}{P_0} \cdot 10^4_{bp} \quad (3.12)$$

Benchmark Price Performance

Benchmark price performance measures are the simplest of the TCA performance evaluation techniques. These are intended to compare specific measures such as net difference and tracking error, or to distinguish between temporary and permanent impact. Some of the more commonly used benchmark prices include:

- Open—as a proxy for arrival price.
- Close—insight into end-of-day tracking error and is more commonly used by index funds that use the closing price in valuation of the fund.
- Next Day Open—as a way to distinguish between temporary and permanent market impact.
- Next Day Close or Future Day Close—also to distinguish between temporary and permanent impact.

The benchmark PnL calculation is:

$$\text{Trading PnL}_{bp} = -1 \cdot \text{Side} \cdot \frac{P_{avg} - P_B}{P_B} \cdot 10^4_{bp} \quad (3.13)$$

where P_B = benchmark price.

VWAP Benchmark

The VWAP benchmark is used as a proxy for fair market price. It helps investors determine if their execution prices were in line and consistent with fair market prices.

The calculation is:

$$\text{VWAP PnL}_{bp} = -1 \cdot \text{Side} \cdot \frac{P_{avg} - \text{VWAP}}{\text{VWAP}} \cdot 10^4_{bp} \quad (3.14)$$